

OEM REFERENCE GUIDE – CENTRIFUGAL PUMPS

Equipment Class: Centrifugal Pump (applicable to P-101, P-204, P-156, P-089, and similar plant pump assets)

Compiled from general industry maintenance practice for centrifugal pump equipment.

SECTION 1 – SAFETY PRECAUTIONS BEFORE INSPECTION

Before any inspection or maintenance work, power to the pump must be disconnected and locked out per standard lock-out/tag-out procedure. Suction and discharge valves should be closed, the pump drained, and any hazardous residue neutralized before opening the casing.

SECTION 2 – ROUTINE INSPECTION SCHEDULE

- Daily: Visual walkaround for leakage at pipe joints and mounting brackets; listen for abnormal noise suggestive of cavitation (a rattling or "marbles in the pump" sound) or high-pitched bearing squeal.
- Weekly: Log discharge pressure and flow rate. A drop of roughly 10% or more from baseline typically indicates internal wear, impeller damage, or suction-side clogging.
- Monthly: Inspect and clean motor cooling fins; debris accumulation is a common cause of heat-related bearing failure.
- Quarterly: Check foundation and mounting bolts, and piping supports, for vibration-induced loosening.

SECTION 3 – MECHANICAL SEAL CONDITION AND FAILURE PROGRESSION

Mechanical seals should be checked via the housing's inspection point:

- Dry condition: seal is functioning normally.
- Intermittent dripping: early indication of seal face wear; maintenance should be scheduled soon, not treated as an emergency yet.
- Continuous/steady leakage: seal has effectively failed; replacement is required immediately, before fluid reaches and damages the motor bearings.

Common root causes of premature seal failure include:

- Running the pump dry, even briefly, which is one of the most damaging conditions for mechanical seals.
- Operating fluid temperature or chemical composition outside the seal material's rated tolerance.
- Abrasive solids settling near the seal during idle/shutdown periods ("abrasive lag"), often prevented by flushing the pump before extended downtime.

SECTION 4 ?? IMPELLER AND CASING INSPECTION

During any teardown or seal replacement, the impeller and casing (volute) should also be inspected for:

- Erosion or pitting on impeller vanes (a honeycomb-like surface pattern), typically caused by cavitation or abrasive fluid content.
- Excessive clearance between impeller and casing due to wear, which reduces pump efficiency and discharge head.
- Debris lodged in impeller passages, which can cause vibration and hydraulic imbalance.

SECTION 5 ?? COMMON FAILURE MODES AND LIKELY CAUSES (troubleshooting reference)

- Excessive heat: commonly caused by dead-heading (running against a closed discharge valve) or bearing friction from inadequate lubrication.
- Rapid/repeated seal wear: often linked to abrasive lag during shutdowns, or seal material incompatible with process fluid chemistry.
- Cavitation noise/vibration: usually indicates insufficient Net Positive Suction Head (NPSH) available, a clogged suction strainer, or a suction-side restriction.

SECTION 6 ?? EXTENDED SHUTDOWN / STORAGE GUIDANCE

If a pump will be out of service for more than approximately 30 days:

1. Drain and flush all process fluid from the casing.
2. Apply a light protective oil coating to internal cast iron components to prevent rust-binding.
3. Manually rotate the shaft by hand periodically (roughly monthly) to prevent seal faces from bonding and to redistribute bearing lubrication.

DATA PROVENANCE NOTE

This document is a synthesized reference compiled from general, publicly available centrifugal pump maintenance guidance (multiple manufacturer service manuals and industry maintenance references). It represents general engineering practice for this equipment class, not a manufacturer-specific document for any single named OEM. It is intended to give the Maintenance Intelligence & RCA Agent a real technical grounding for failure-mode reasoning about pump equipment (e.g., P-204's recurring seal failure pattern), rather than relying solely on synthetic work-order text.